

Math 104 – Lecture 03

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1 Glossary

- $A_N = \{s_n : n \geq N\}$: the *tail* of the sequence, with the first $N - 1$ terms removed.
- $v_N = \inf(A_N)$ and $u_N = \sup(A_N)$: the lower and upper edges of the tail. As the tail shrinks, v_N increases and u_N decreases.
- $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$: the extended reals, in which every monotone sequence has a limit.
- $s_n \rightarrow L$ means that for every $\varepsilon > 0$, there exists N such that $|s_n - L| < \varepsilon$ for all $n \geq N$.

2 Recall: $\limsup$ and $\liminf$

Let $(s_n) \subset \mathbb{R}$ and

$$A_N = \{s_n : n \geq N\}.$$

Since $v_N = \inf(A_N)$ increases and $u_N = \sup(A_N)$ decreases, both converge in $\overline{\mathbb{R}}$. We define

$$\liminf s_n = \lim_{N \rightarrow \infty} \inf(A_N), \quad \limsup s_n = \lim_{N \rightarrow \infty} \sup(A_N).$$

Always,

$$\liminf s_n \leq \limsup s_n.$$

Intuitively, $\limsup s_n$ is the upper edge that the sequence approaches, while $\liminf s_n$ is the lower edge.

For example, if $s_n = (-1)^n$, every tail contains both 1 and -1 , so

$$\limsup s_n = 1, \quad \liminf s_n = -1.$$

So when does the sequence actually converge?

3 Convergence

Theorem 3.1 *Let $(s_n) \subset \mathbb{R}$. Then (s_n) converges to some $L \in \mathbb{R}$ if and only if*

$$\liminf s_n = \limsup s_n.$$

($\implies$) Suppose $s_n \rightarrow L$. Given $\varepsilon > 0$, choose N such that

$$|s_n - L| < \varepsilon \quad \text{for all } n \geq N.$$

Then every element of A_N lies in $(L - \varepsilon, L + \varepsilon)$, so

$$L - \varepsilon \leq v_N \leq \liminf s_n \leq \limsup s_n \leq u_N \leq L + \varepsilon.$$

Since this holds for every $\varepsilon > 0$,

$$\liminf s_n = \limsup s_n = L.$$

($\impliedby$) Suppose

$$\liminf s_n = \limsup s_n = L \in \mathbb{R}.$$

Given $\varepsilon > 0$, since $v_n \uparrow L$ and $u_n \downarrow L$, there exist N_1, N_2 such that

$$v_{N_1} > L - \varepsilon, \quad u_{N_2} < L + \varepsilon.$$

Let

$$N = \max(N_1, N_2).$$

Then for every $n \geq N$,

$$L - \varepsilon < v_n \leq s_n \leq u_n < L + \varepsilon.$$

Hence

$$|s_n - L| < \varepsilon$$

for all $n \geq N$, so

$$s_n \rightarrow L.$$

4 Cauchy Sequence

A sequence $s(n) \subset \mathbb{R}$ is Cauchy is

$$\forall \epsilon > 0, \exists N > 0 \text{ s.t. } m, n > N \rightarrow |s_m - s_n| < \epsilon$$

Lemma 4.1 (Convergence) *Every convergent sequence is Cauchy*

Proof. Let $(s_n) \subset \mathbb{R}$ convergent. Then suppose we have $\epsilon > 0$ For $\frac{\epsilon}{2}$,

$$\exists N > 0 \text{ s.t } n > N \implies |s_n - L| < \frac{\epsilon}{2}$$

so its true. (just plug it in)

□

5 Definitions and notation

Definition 5.1 (First definition). Write the definition and one sentence explaining why it matters.

6 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

7 Examples and calculations

Example 7.1 (Lecture example). Write the setup, the calculation, and the conclusion.

8 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: